

CERTIFICATE OF ANALYSIS

Description : Coconut Shell Charcoal Briquette
Type of Sample : Cube, Size (2.5 x 2.5 x 2.5 cm)
Manufacturer / Shipper : CV. Alfa Madya Group
Jalan Zebra Mukti No. 4, Pedurungan Kidul
Semarang, Jawa Tengah, Indonesia 50192
Sample Weight : 3.00 kgs
Date of Sampling : 15 May 2023
Location of Sampling : CV. Alfa Madya Group, Warehouse
Date of Analysis : 16 May 2023
Invoice No. : 01/ALMA/IPL-MAH/V/2023
Booking No. : EBKG05703690
Gross Weight : 18,900 Kgs

Sampling

Samples were taken and prepared in accordance with SNI 19-0428-1998.

Analysis

Analysis were performed at PT Carsurin Semarang Laboratory (accreditation by KAN / National Accreditation Committee – Indonesia for ISO 17025:2017 Cert No. LP-415-IDN) in accordance with ASTM Standard. The results of our analysis are as follows:

Parameter	Unit	Results		Method
		Wet Basis	Dry Basis	
Moisture	Pct	5.91	-	ASTM D1762-84 (REAPPROVED 2021)
Ash Content	Pct	2.56	2.72	ASTM D1762-84 (REAPPROVED 2021)
Volatile Matter	Pct	16.48	17.52	ASTM D1762-84 (REAPPROVED 2021)
Fixed Carbon	Pct	75.05	79.76	By Calculation

This certificate refers solely to the analysis in accordance with the details described herein and does not certify any other matter, and is issued without prejudice.

Jakarta, 19 May 2023

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Dede Jemmi Supriadi

Head of Laboratory

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The analysis results are based on representative samples which are taken based on applied standard (Ref SNI 19-0428-1998 *Petunjuk Pengambilan Contoh Padatan*) during the inspection (or audit). This particular cargo as stated in this certificate has been performed cooling and weathering process as declared by the shipper in the weathering certificate. The certificate is valid for this particular shipment with the invoice number, booking number, and gross/net weight as mentioned in this certificate.



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SUBJECT

Evaluation of "Coconut Shell Charcoal Briquette" Sample for Self-Heating Substance Test

MANUFACTURER / SHIPPER

CV. Alfa Madya Group
Jalan Zebra Mukti No. 4, Pedurungan Kidul
Semarang, Jawa Tengah, Indonesia 50192

DATE OF SAMPLING

15 May 2023

LOCATION OF SAMPLING

CV. Alfa Madya Group, Warehouse

DESCRIPTION & TYPE OF SAMPLE

Coconut Shell Charcoal Briquette, Cube, Size (2.5 x 2.5 x 2.5 cm)

DATE OF TESTING

16 May 2023 up to 18 May 2023

INVOICE NO.

01/ALMA/IPL-MAH/V/2023

BOOKING NO.

EBKG05703690

GROSS WEIGHT

18,900 Kgs

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METHOD OF TEST

The self heating substances test was performed according to United Nations "Recommendations on the Transport Dangerous Good, Manual of Test and Criteria" (33.4.3.3.1) as per revised UN test manual 7th edition 2019. Self heating substances, using method of UN 33.4.6 "Test N.4: Test method self-heating substances".

A consignment of carbon if it passes the tests for self-heating substances as reflected in the United Nations Manual of Tests and Criteria (see 33.4.3.3), and is accompanied by a certificate from a laboratory accredited by the competent authority, stating that the product to be loaded has been correctly sampled by trained staff from that laboratory and that the sample was correctly tested and has passed the test.

The sample received was placed into a 100 mm cubic mesh sample container made of stainless steel. The container was then hung at the centre of the oven. The oven temperature was slowly raised to 140° C and kept constant for 24 hours. The temperature at the centre of the sample and temperature between the container and oven were recorded continuously.

RESULT

Table 1: Analysis result for "Coconut Shell Charcoal Briquette" sample

Sample	Oven Temperature (°C)	Cube Size (mm)	Maximum Temperature Reached (°C)	Result
Coconut Shell Charcoal Briquette	140	100	200.9	Positive
Coconut Shell Charcoal Briquette	140	25	140.4	Negative
Coconut Shell Charcoal Briquette	120	100	123.4	Negative

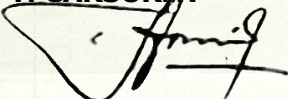
The above result showed that the "Coconut Shell Charcoal Briquette" sample shall not be classified in self heating substances and the substance is to be contained in packages of not more than 3 m³.

Remark

The temperature profiles were enclosed in next pages for your references. T1 represented temperature in the center of the sample. T2 represented temperature between the container and the oven wall.

Jakarta, 19 May 2023

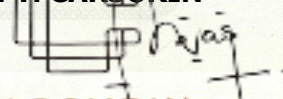
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Dede Jemmi Subriadi
Head of Laboratory

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Aniran Lesmana
Head of Technical Charcoal



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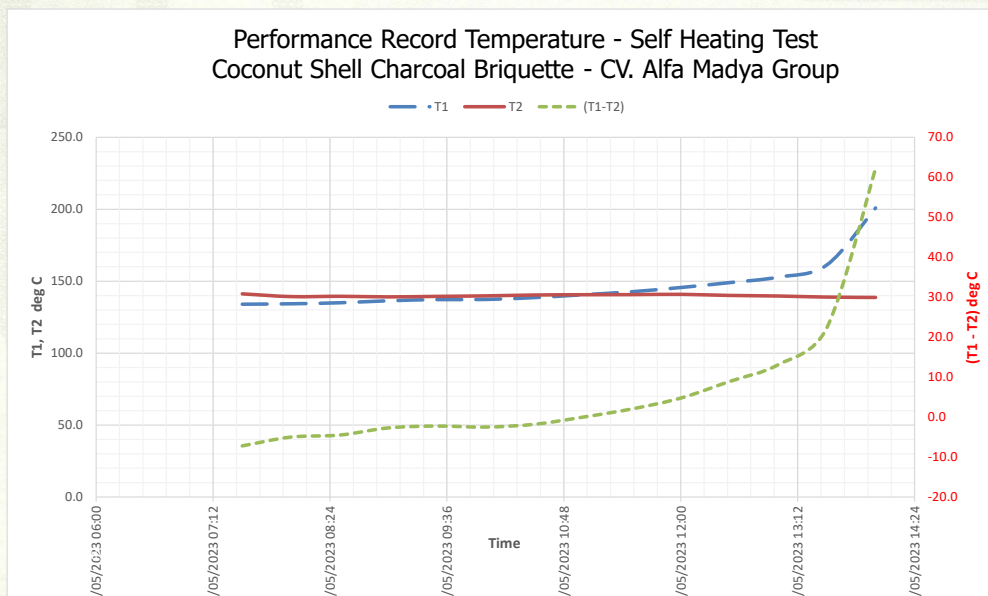
Temperature Profile "Coconut Shell Charcoal Briquette" at 140⁰ C (100mm cube)

Start Time : 16-May-23 07:30:00
 Stop Time : 16-May-23 14:00:00
 Elapsed Time : 6:30:00
 Interval : 00:30:00
 Total Readings : 14
 Thermocouple Type : K
 Scaling : (None)

Temperature	Max Time Stamp	Max	Average	Min	Min Time Stamp
T1	16/05/2023 14:00	200.9	146.1	134.0	16/05/2023 07:30
T2	16/05/2023 07:30	141.2	139.9	138.7	16/05/2023 14:00
T1-T2	16/05/2023 14:00	62.2	6.2	-7.2	16/05/2023 07:30

Remark:

T1 represented temperature in the centre of the sample. T2 represented temperature between the container and the oven wall.



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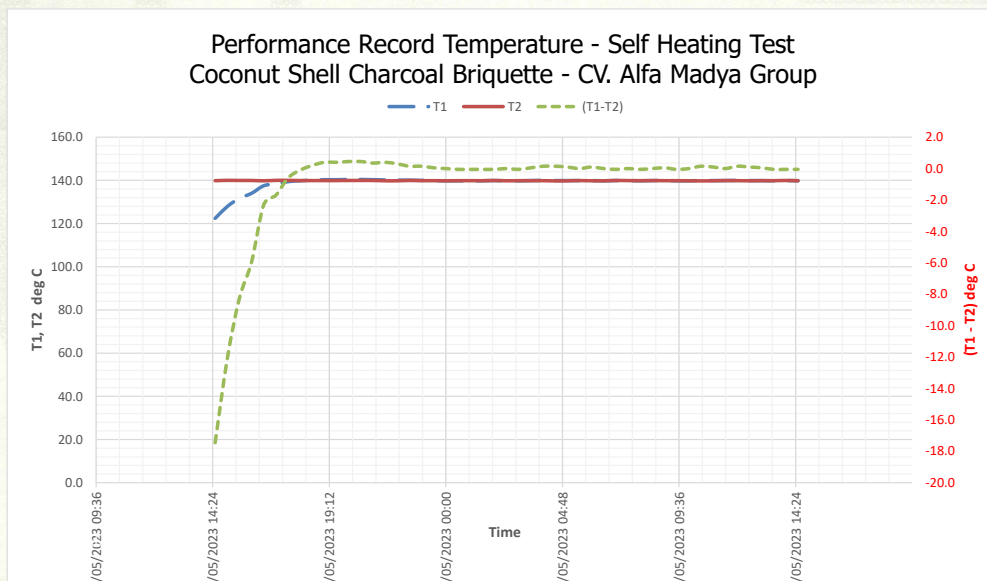
24 Hours Temperature Profile "Coconut Shell Charcoal Briquette" at 140⁰ C (25mm cube)

Start Time : 16-May-23 14:30:00
 Stop Time : 17-May-23 14:30:00
 Elapsed Time : 24:00:00
 Interval : 00:30:00
 Total Readings : 49
 Thermocouple Type : K
 Scaling : (None)

Temperature	Max Time Stamp	Max	Average	Min	Min Time Stamp
T1	16/05/2023 20:00	140.4	139.0	122.4	16/05/2023 14:30
T2	16/05/2023 15:00	140.0	139.9	139.8	16/05/2023 16:30
T1-T2	16/05/2023 20:00	0.5	-0.9	-17.5	16/05/2023 14:30

Remark:

T1 represented temperature in the centre of the sample. T2 represented temperature between the container and the oven wall.



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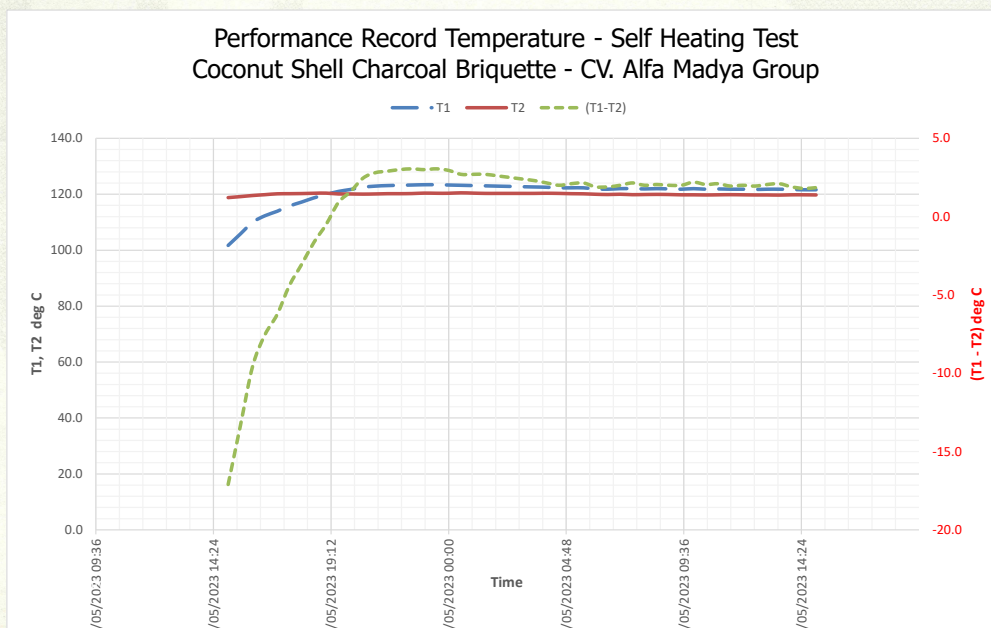
24 Hours Temperature Profile "Coconut Shell Charcoal Briquette" at 120⁰ C (100mm cube)

Start Time : 17-May-23 15:00:00
 Stop Time : 18-May-23 15:00:00
 Elapsed Time : 24:00:00
 Interval : 00:30:00
 Total Readings : 49
 Thermocouple Type : K
 Scaling : (None)

Temperature	Max Time Stamp	Max	Average	Min	Min Time Stamp
T1	17/05/2023 23:00	123.4	120.6	101.7	17/05/2023 15:00
T2	18/05/2023 00:30	120.5	120.0	118.8	17/05/2023 15:00
T1-T2	17/05/2023 22:30	3.1	0.5	-17.1	17/05/2023 15:00

Remark:

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